ZEARN Math

Minnesota State Standards Alignment

For Grades K-8

About Zearn

Zearn is the 501(c)(3) nonprofit educational organization behind Zearn Math, the top-rated math learning platform used by 1 in 4 elementary-school students and by more than 1 million middle-school students nationwide. Zearn Math supports teachers with research-backed curriculum and digital lessons proven to double student academic growth across all levels of student proficiency. Zearn Math can be used flexibly to support a range of instructional needs including Tier 1 and 2 support, acceleration, intervention, tutoring, after school programming and summer programming.

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Zearn Math is aligned to Minnesota's Academic Standards in Mathematics.

We are committed to ensuring that our high-quality instructional materials align to Minnesota's Academic Standards in Mathematics to fully support Minnesota's students and teachers. The tables on the following pages, organized by Minnesota Academic Standards, identify the alignment between each grade-level standard and Zearn Math lesson(s).

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Kindergarten

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
Kindergarten Standards		Lessons
Number & Operation		
K.1.1.1	Recognize that a number can be used to represent how many objects are in a set or to represent the position of an object in a sequence.	Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 1, Topic G Mission 1, Topic H Mission 3, Topic B Mission 3, Topic F Mission 3, Topic G Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E
K.1.1.2	Read, write, and represent whole numbers from 0 to at least 31. Representations may include numerals, pictures, real objects and picture graphs, spoken words, and manipulatives such as connecting cubes.	Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E
K.1.1.3	Count, with and without objects, forward and backward to at least 20.	Mission 1, Topic G Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D
K.1.1.4	Find a number that is 1 more or 1 less than a given number.	Mission 1, Topic G Mission 1, Topic H
K.1.1.5	Compare and order whole numbers, with and without objects, from 0 to 20.	Mission 3, Topic F Mission 3, Topic G Mission 3, Topic H

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
Kindergarten Standards		**Lessons**
K.1.2.1	Use objects and draw pictures to find the sums and differences of numbers between 0 and 10.	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D Mission 4, Topic G Mission 4, Topic H
K.1.2.2	Compose and decompose numbers up to 10 with objects and pictures.	Mission 1, Topic C Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D Mission 4, Topic E Mission 4, Topic G Mission 4, Topic H
Algebra		
K.2.1.1	Identify, create, complete, and extend simple patterns using shape, color, size, number, sounds and movements. Patterns may be repeating, growing or shrinking such as ABB, ABB, ABB or ●, ●●, ●●●.	Mission 1, Topic H Mission 4, Topic B Mission 4, Topic H Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C
Geometry & Measurement		
K.3.1.1	Recognize basic two- and three-dimensional shapes such as squares, circles, triangles, rectangles, trapezoids, hexagons, cubes, cones, cylinders and spheres.	Mission 2, Topic C
K.3.1.2	Sort objects using characteristics such as shape, size, color and thickness.	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 6, Topic A Mission 6, Topic B
K.3.1.3	Use basic shapes and spatial reasoning to model objects in the real-world.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic E Mission 2, Topic A

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
Kindergarten Standards		Lessons
		Mission 2, Topic B Mission 2, Topic C Mission 6, Topic A Mission 6, Topic B
K.3.2.1	Use words to compare objects according to length, size, weight and position.	MIssion 3, Topic A MIssion 3, Topic B MIssion 3, Topic C MIssion 3, Topic D MIssion 3, Topic F MIssion 3, Topic H
K.3.2.2	Order two or three objects using measurable attributes, such as length and weight.	MIssion 3, Topic A MIssion 3, Topic B MIssion 3, Topic C MIssion 3, Topic D MIssion 3, Topic F MIssion 3, Topic H

1st Grade

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
1st Grade Standards		Lessons
Number & Operation		
1.1.1.1	Use place value to describe whole numbers between 10 and 100 in terms of tens and ones.	Mission 2, Topic D Mission 4, Topic A Mission 4, Topic B Mission 4, Topic F Mission 6, Topic B
1.1.1.2	Read, write and represent whole numbers up to 120. Representations may include numerals, addition and subtraction, pictures, tally marks, number lines and manipulatives, such as bundles of sticks and base 10 blocks.	Mission 4, Topic A Mission 6, Topic B
1.1.1.3	Count, with and without objects, forward and backward from any given number up to 120.	Mission 4, Topic A Mission 6, Topic B
1.1.1.4	Find a number that is 10 more or 10 less than a given number.	Mission 2, Topic D Mission 4, Topic A Mission 4, Topic C Mission 6, Topic B Mission 6, Topic C
1.1.1.5	Compare and order whole numbers up to 120.	Mission 4, Topic B Mission 6, Topic B
1.1.1.6	Use words to describe the relative size of numbers.	Mission 4, Topic B Mission 6, Topic B
1.1.1.7	Use counting and comparison skills to create and analyze bar graphs and tally charts.	Grade 2, Mission 7, Topic A
1.1.2.1	Use words, pictures, objects, length-based models (connecting cubes), numerals and number lines to model and solve addition and subtraction problems in part-part-total, adding to, taking away from and comparing situations.	Mission 4, Topic C Mission 4, Topic D Mission 4, Topic F Mission 6, Topic C Mission 6, Topic D

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
1st Grade Standards		Lessons
1.1.2.2	Compose and decompose numbers up to 12 with an emphasis on making ten.	Mission 2, Topic D Mission 4, Topic A Mission 4, Topic B Mission 4, Topic F Mission 6, Topic B
1.1.2.3	Recognize the relationship between counting and addition and subtraction. Skip count by 2s, 5s, and 10s.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 1, Topic G Mission 1, Topic H Mission 1, Topic I Mission 2, Topic B Mission 2, Topic C Grade 2, Mission 6, Topic D
Algebra		
1.2.1.1	Create simple patterns using objects, pictures, numbers and rules. Identify possible rules to complete or extend patterns. Patterns may be repeating, growing or shrinking. Calculators can be used to create and explore patterns.	Mission 1, Topic B Mission 1, Topic D Mission 1, Topic F Mission 1, Topic J Mission 2, Topic A Mission 2, Topic B Mission 4, Topic D Mission 6, Topic B
1.2.2.1	Represent real-world situations involving addition and subtraction basic facts, using objects and number sentences.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic G Mission 1, Topic H Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D Mission 3, Topic C Mission 3, Topic D Mission 4, Topic E Mission 6, Topic A Mission 6, Topic F

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
1st Grade Standards		Lessons
1.2.2.2	Determine if equations involving addition and subtraction are true.	Mission 1, Topic E Mission 2, Topic B Mission 2, Topic C
1.2.2.3	Use number sense and models of addition and subtraction, such as objects and number lines, to identify the missing number in an equation such as: 2 + 4 = __ 3 + __ = 7 5 = __ – 3.	Mission 1, Topic D Mission 2, Topic H Mission 2, Topic C
1.2.2.4	Use addition or subtraction basic facts to represent a given problem situation using a number sentence.	Mission 1, Topic B Mission 1, Topic D Mission 1, Topic F Mission 1, Topic J Mission 2, Topic A Mission 2, Topic B Mission 4, Topic D Mission 6, Topic B
Geometry & Measurement		
1.3.1.1	Describe characteristics of two- and three-dimensional objects, such as triangles, squares, rectangles, circles, rectangular prisms, cylinders, cones and spheres.	Mission 5, Topic A
1.3.1.2	Compose (combine) and decompose (take apart) two- and threedimensional figures such as triangles, squares, rectangles, circles, rectangular prisms and cylinders.	Mission 5, Topic B
1.3.2.1	Measure the length of an object in terms of multiple copies of another object.	Mission 3, Topic B Mission 3, Topic C
1.3.2.2	Tell time to the hour and half-hour.	Mission 5, Topic D
1.3.2.3	Identify pennies, nickels and dimes; find the value of a group of these coins, up to one dollar.	Mission 4, Topic A Mission 6, Topic E

2nd Grade

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
Number & Operation		
2.1.1.1	Read, write and represent whole numbers up to 1000. Representations may include numerals, addition, subtraction, multiplication, words, pictures, tally marks, number lines and manipulatives, such as bundles of sticks and base 10 blocks.	Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E Mission 3, Topic G Mission 6, Topic A Mission 6, Topic B Mission 7, Topic B Mission 7, Topic E Mission 8, Topic D
2.1.1.2	Use place value to describe whole numbers between 10 and 1000 in terms of hundreds, tens and ones. Know that 100 is 10 tens, and 1000 is 10 hundreds.	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
2.1.1.3	Find 10 more or 10 less than a given three-digit number. Find 100 more or 100 less than a given three-digit number.	Mission 3, Topic G Mission 5, Topic A Mission 5, Topic D
2.1.1.4	Round numbers up to the nearest 10 and 100 and round numbers down to the nearest 10 and 100.	Grade 3, Mission 2, Topic C
2.1.1.5	Compare and order whole numbers up to 1000.	Mission 3, Topic E Mission 3, Topic F
2.1.2.1	Use strategies to generate addition and subtraction facts including making tens, fact families, doubles plus or minus one, counting on, counting back, and the commutative and associative properties. Use the relationship between addition and subtraction to generate basic facts.	Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D Mission 4, Topic E Mission 4, Topic F Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
2nd Grade Standards		**Lessons**
2.1.2.2	Demonstrate fluency with basic addition facts and related subtraction facts.	Mission 1, Topic A Mission 2, Topic B
2.1.2.3	Estimate sums and differences up to 100.	*It is recommended that teachers use supplemental materials to fully address this standard.*
2.1.2.4	Use mental strategies and algorithms based on knowledge of place value and equality to add and subtract two-digit numbers. Strategies may include decomposition, expanded notation, and partial sums and differences.	Mission 1, Topic B Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 7, Topic B Mission 7, Topic E Mission 8, Topic D
2.1.2.5	Solve real-world and mathematical addition and subtraction problems involving whole numbers with up to 2 digits.	Mission 1, Topic B Mission 3, Topic G Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic F
2.1.2.6	Use addition and subtraction to create and obtain information from tables, bar graphs and tally charts.	Mission 7, Topic A
Algebra		
2.2.1.1	Identify, create and describe simple number patterns involving repeated addition or subtraction, skip counting and arrays of objects such as counters or tiles. Use patterns to solve problems in various contexts.	Mission 1, Topic B Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E Mission 3, Topic G Mission 4, Topic A Mission 4, Topic B Mission 4, Topic D Mission 5, Topic A Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C Mission 6, Topic D Mission 7, Topic B Mission 7, Topic E Mission 8, Topic D

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
2nd Grade Standards		**Lessons**
2.2.2.1	Understand how to interpret number sentences involving addition, subtraction and unknowns represented by letters. Use objects and number lines and create real-world situations to represent number sentences.	Mission 1, Topic B Mission 3, Topic G Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic F
2.2.2.2	Use number sentences involving addition, subtraction, and unknowns to represent given problem situations. Use number sense and properties of addition and subtraction to find values for the unknowns that make the number sentences true.	Mission 1, Topic B Mission 3, Topic G Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic F
Geometry & Measurement		
2.3.1.1	Describe, compare, and classify two- and three-dimensional figures according to number and shape of faces, and the number of sides, edges and vertices (corners).	Mission 8, Topic A Mission 8, Topic B Mission 8, Topic C
2.3.1.2	Identify and name basic two- and three-dimensional shapes, such as squares, circles, triangles, rectangles, trapezoids, hexagons, cubes, rectangular prisms, cones, cylinders and spheres.	Mission 8, Topic A Mission 8, Topic B Mission 8, Topic C
2.3.2.1	Understand the relationship between the size of the unit of measurement and the number of units needed to measure the length of an object.	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D Mission 7, Topic C Mission 7, Topic D Mission 7, Topic F Mission 8, Topic A
2.3.2.2	Demonstrate an understanding of the relationship between length and the numbers on a ruler by using a ruler to measure lengths to the nearest centimeter or inch.	Mission 7, Topic A Mission 7, Topic E Mission 7, Topic F
2.3.3.1	Tell time to the quarter-hour and distinguish between a.m. and p.m.	Mission 8, Topic D

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
2.3.3.2	Identify pennies, nickels, dimes and quarters. Find the value of a group of coins and determine combinations of coins that equal a given amount.	Mission 3, Topic D Mission 7, Topic B

3rd Grade

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
Number & Operation		
3.1.1.1	Read, write and represent whole numbers up to 100,000. Representations may include numerals, expressions with operations, words, pictures, number lines, and manipulatives such as bundles of sticks and base 10 blocks.	Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E
3.1.1.2	Use place value to describe whole numbers between 1000 and 100,000 in terms of ten thousands, thousands, hundreds, tens and ones.	Grade 4, Mission 1, Topic A
3.1.1.3	Find 10,000 more or 10,000 less than a given five-digit number. Find 1000 more or 1000 less than a given four- or five-digit. Find 100 more or 100 less than a given four- or five-digit number.	Grade 2, Mission 5, Topic A
3.1.1.4	Round numbers to the nearest 10,000, 1,000, 100 and 10. Round up and round down to estimate sums and differences.	Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E Grade 4, Mission 1, Topic C
3.1.1.5	Compare and order whole numbers up to 100,000.	Grade 4, Mission 1, Topic B
3.1.2.1	Add and subtract multi-digit numbers, using efficient and generalizable procedures based on knowledge of place value, including standard algorithms.	Mission 2, Topic A Mission 2, Topic D Mission 2, Topic E
3.1.2.2	Use addition and subtraction to solve real-world and mathematical problems involving whole numbers. Use various strategies, including the relationship between addition and subtraction, the use of technology, and the context of the problem to assess the reasonableness of results.	Mission 1, Topic F Mission 2, Topic D Mission 2, Topic E Mission 3, Topic C Mission 3, Topic E Mission 3, Topic F Mission 7, Topic A
3.1.2.3	Represent multiplication facts by using a variety of approaches, such as repeated addition, equal-sized groups, arrays, area models, equal jumps on a number	Mission 1, Topic C Mission 1, Topic E Mission 1, Topic F

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
3rd Grade Standards		**Lessons**
	line and skip counting. Represent division facts by using a variety of approaches, such as repeated subtraction, equal sharing and forming equal groups. Recognize the relationship between multiplication and division.	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic F
3.1.2.4	Solve real-world and mathematical problems involving multiplication and division, including both "how many in each group" and "how many groups" division problems.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
3.1.2.5	Use strategies and algorithms based on knowledge of place value, equality and properties of addition and multiplication to multiply a two or three-digit number by a one-digit number. Strategies may include mental strategies, partial products, the standard algorithm, and the commutative, associative, and distributive properties.	Grade 4, Mission 3, Topic B Grade 4, Mission 3, Topic C Grade 4, Mission 3, Topic D Grade 4, Mission 3, Topic H
3.1.3.1	Read and write fractions with words and symbols. Recognize that fractions can be used to represent parts of a whole, parts of a set, points on a number line, or distances on a number line.	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C
3.1.3.2	Understand that the size of a fractional part is relative to the size of the whole.	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E Mission 5, Topic F
3.1.3.3	Order and compare unit fractions and fractions with like denominators by using models and an understanding of the concept of numerator and denominator.	Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E Mission 5, Topic F

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
Algebra		
3.2.1.1	Create, describe, and apply single-operation input-output rules involving addition, subtraction and multiplication to solve problems in various contexts.	Mission 3, Topic A Mission 3, Topic D Mission 3, Topic E Mission 3, Topic F
3.2.2.1	Understand how to interpret number sentences involving multiplication and division basic facts and unknowns. Create real-world situations to represent number sentences.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E Mission 3, Topic F
3.2.2.2	Use multiplication and division basic facts to represent a given problem situation using a number sentence. Use number sense and multiplication and division basic facts to find values for the unknowns that make the number sentences true.	Mission 1, Topic A Mission 1, Topic C Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E Mission 3, Topic F
Geometry & Measurement		
3.3.1.1	Identify parallel and perpendicular lines in various contexts, and use them to describe and create geometric shapes, such as right triangles, rectangles, parallelograms and trapezoids.	Grade 4, Mission 4 Topic D
3.3.1.2	Sketch polygons with a given number of sides or vertices (corners), such as pentagons, hexagons and octagons.	Mission 7, Topic B

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
3rd Grade Standards		**Lessons**
3.3.2.1	Use half units when measuring distances.	Mission 6, Topic B Mission 7, Topic D
3.3.2.2	Find the perimeter of a polygon by adding the lengths of the sides.	Mission 7, Topic C Mission 7, Topic D Mission 7, Topic E
3.3.2.3	Measure distances around objects.	Mission 7, Topic C
3.3.3.1	Tell time to the minute, using digital and analog clocks. Determine elapsed time to the minute.	Mission 2, Topic A Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E
3.3.3.2	Know relationships among units of time.	Mission 2, Topic A Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E
3.3.3.3	Make change up to one dollar in several different ways, including with as few coins as possible.	Grade 2, Mission 7, Topic B
3.3.3.4	Use an analog thermometer to determine temperature to the nearest degree in Fahrenheit and Celsius.	*It is recommended that teachers use supplemental materials to fully address this standard.*
Data Analysis		
3.4.1.1	Collect, display and interpret data using frequency tables, bar graphs, picture graphs and number line plots having a variety of scales. Use appropriate titles, labels and units.	Mission 6, Topic A

4th Grade

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
Number & Operation		
4.1.1.1	Demonstrate fluency with multiplication and division facts.	Grade 3, Mission 3, Topic A Grade 3, Mission 3, Topic B Grade 3, Mission 3, Topic C Grade 3, Mission 3, Topic D Grade 3, Mission 3, Topic E
4.1.1.2	Use an understanding of place value to multiply a number by 10, 100 and 1000.	Mission 1, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic G
4.1.1.3	Multiply multi-digit numbers, using efficient and generalizable procedures, based on knowledge of place value, including standard algorithms.	Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic H
4.1.1.4	Estimate products and quotients of multi-digit whole numbers by using rounding, benchmarks and place value to assess the reasonableness of results.	Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic D Mission 3, Topic E Mission 3, Topic G Mission 3, Topic H Mission 7, Topic B Mission 7, Topic C
4.1.1.5	Solve multi-step real-world and mathematical problems requiring the use of addition, subtraction and multiplication of multi-digit whole numbers. Use various strategies, including the relationship between operations, the use of technology, and the context of the problem to assess the reasonableness of results.	Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic D Mission 3, Topic E Mission 3, Topic G Mission 3, Topic H Mission 7, Topic B Mission 7, Topic C

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
4th Grade Standards		**Lessons**
4.1.1.6	Use strategies and algorithms based on knowledge of place value, equality and properties of operations to divide multi-digit whole numbers by one- or two-digit numbers. Strategies may include mental strategies, partial quotients, the commutative, associative, and distributive properties and repeated subtraction.	Mission 3, Topic E Mission 3, Topic F Mission 3, Topic G
4.1.2.1	Represent equivalent fractions using fraction models such as parts of a set, fraction circles, fraction strips, number lines and other manipulatives. Use the models to determine equivalent fractions.	Mission 1, Topic B Mission 7, Topic B Mission 7, Topic C
4.1.2.2	Locate fractions on a number line. Use models to order and compare whole numbers and fractions, including mixed numbers and improper fractions.	Mission 5, Topic C Mission 5, Topic E
4.1.2.3	Use fraction models to add and subtract fractions with like denominators in real-world and mathematical situations. Develop a rule for addition and subtraction of fractions with like denominators.	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E Mission 5, Topic F Mission 6, Topic D
4.1.2.4	Read and write decimals with words and symbols; use place value to describe decimals in terms of thousands, hundreds, tens, ones, tenths, hundredths and thousandths.	Mission 6, Topic A Mission 6, Topic B Grade 5, Mission 1, Topic B
4.1.2.5	Compare and order decimals and whole numbers using place value, a number line and models such as grids and base 10 blocks.	Mission 6, Topic B Mission 6, Topic C
4.1.2.6	Read and write tenths and hundredths in decimal and fraction notations using words and symbols; know the fraction and decimal equivalents for halves and fourths.	Mission 6, Topic A Mission 6, Topic B Mission 6, Topic D Mission 6, Topic E
4.1.2.7	Round decimals to the nearest tenth.	Grade 5, Mission 1, Topic C
Algebra		

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
4th Grade Standards		**Lessons**
4.2.1.1	Create and use input-output rules involving addition, subtraction, multiplication and division to solve problems in various contexts. Record the inputs and outputs in a chart or table.	Mission 5, Topic H Grade 5, Mission 6, Topic B Grade 5, Mission 6, Topic D
4.2.2.1	Understand how to interpret number sentences involving multiplication, division and unknowns. Use real-world situations involving multiplication or division to represent number sentences.	Mission 1, Topic A Mission 3, Topic A Mission 3, Topic B Mission 3, Topic D Mission 7, Topic A
4.2.2.2	Use multiplication, division and unknowns to represent a given problem situation using a number sentence. Use number sense, properties of multiplication, and the relationship between multiplication and division to find values for the unknowns that make the number sentences true.	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 5, Topic G Mission 7, Topic A Mission 7, Topic B
Geometry & Measurement		
4.3.1.1	Describe, classify and sketch triangles, including equilateral, right, obtuse and acute triangles. Recognize triangles in various contexts.	Mission 4, Topic D
4.3.1.2	Describe, classify and draw quadrilaterals, including squares, rectangles, trapezoids, rhombuses, parallelograms and kites. Recognize quadrilaterals in various contexts.	Mission 4, Topic D
4.3.2.1	Measure angles in geometric figures and real-world objects with a protractor or angle ruler.	Mission 4, Topic B
4.3.2.2	Compare angles according to size. Classify angles as acute, right and obtuse.	Mission 4, Topic B
4.3.2.3	Understand that the area of a two-dimensional figure can be found by counting the total number of same size square units that cover a shape without gaps or overlaps. Justify why length and width are multiplied to find the area of a rectangle by breaking the rectangle into one unit by one unit squares and viewing these as grouped into rows and columns.	Grade 3, Mission 4, Topic A Grade 3, Mission 4, Topic B Grade 3, Mission 4, Topic C Grade 3, Mission 4, Topic D

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
4.3.2.4	Find the areas of geometric figures and real-world objects that can be divided into rectangular shapes. Use square units to label area measurements.	Mission 3, Topic A Mission 3, Topic H
4.3.3.1	Apply translations (slides) to figures.	Grade 8, Mission 1, Topic A Grade 8, Mission 1, Topic B Grade 8, Mission 1, Topic C
4.3.3.2	Apply reflections (flips) to figures by reflecting over vertical or horizontal lines and relate reflections to lines of symmetry.	Mission 4, Topic D Grade 8, Mission 1, Topic A Grade 8, Mission 1, Topic B Grade 8, Mission 1, Topic C
4.3.3.3	Apply rotations (turns) of 90˚ clockwise or counterclockwise.	Mission 4, Topic B Grade 8, Mission 1, Topic A Grade 8, Mission 1, Topic B Grade 8, Mission 1, Topic C
4.3.3.4	Recognize that translations, reflections and rotations preserve congruency and use them to show that two figures are congruent.	Grade 8, Mission 1, Topic A Grade 8, Mission 1, Topic B Grade 8, Mission 1, Topic C
Data Analysis		
4.4.1.1	Use tables, bar graphs, timelines and Venn diagrams to display data sets. The data may include fractions or decimals. Understand that spreadsheet tables and graphs can be used to display data.	Mission 5, Topic E Mission 5, Topic G

5th Grade

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
Number & Operation		
5.1.1.1	Divide multi-digit numbers, using efficient and generalizable procedures, based on knowledge of place value, including standard algorithms. Recognize that quotients can be represented in a variety of ways, including a whole number with a remainder, a fraction or mixed number, or a decimal.	Mission 2, Topic E Mission 2, Topic F Mission 2, Topic H Mission 3, Topic A Mission 4, Topic B
5.1.1.2	Consider the context in which a problem is situated to select the most useful form of the quotient for the solution and use the context to interpret the quotient appropriately.	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 4, Topic B Mission 4, Topic D
5.1.1.3	Estimate solutions to arithmetic problems in order to assess the reasonableness of results.	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic E Mission 2, Topic F Mission 2, Topic G
5.1.1.4	Solve real-world and mathematical problems requiring addition, subtraction, multiplication and division of multi-digit whole numbers. Use various strategies, including the inverse relationships between operations, the use of technology, and the context of the problem to assess the reasonableness of results.	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic E Mission 2, Topic F
5.1.2.1	Read and write decimals using place value to describe decimals in terms of groups from millionths to millions.	Mission 1, Topic B Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F
5.1.2.2	Find 0.1 more than a number and 0.1 less than a number. Find 0.01 more than a number and 0.01 less than a number. Find 0.001 more than a number and 0.001 less than a number.	Mission 1, Topic D

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
5.1.2.3	Order fractions and decimals, including mixed numbers and improper fractions, and locate on a number line.	Mission 1, Topic B Grade 4, Mission 5, Topic C Grade 4, Mission 5, Topic E
5.1.2.4	Recognize and generate equivalent decimals, fractions, mixed numbers and improper fractions in various contexts.	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D
5.1.2.5	Round numbers to the nearest 0.1, 0.01 and 0.001.	Mission 1, Topic C
5.1.3.1	Add and subtract decimals and fractions, using efficient and generalizable procedures, including standard algorithms.	Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 2, Topic C Mission 2, Topic D Mission 2, Topic G Mission 2, Topic H Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 4, Topic E Mission 4, Topic G
5.1.3.2	Model addition and subtraction of fractions and decimals using a variety of representations.	Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 2, Topic C Mission 2, Topic D Mission 2, Topic G Mission 2, Topic H Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 4, Topic E Mission 4, Topic G
5.1.3.3	Estimate sums and differences of decimals and fractions to assess the reasonableness of results.	Mission 3, Topic D
5.1.3.4	Solve real-world and mathematical problems requiring addition and subtraction of decimals,	Mission 3, Topic B Mission 3, Topic C

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
5th Grade Standards		**Lessons**
	fractions and mixed numbers, including those involving measurement, geometry and data.	Mission 3, Topic D Mission 4, Topic D
Algebra		
5.2.1.1	Create and use rules, tables, spreadsheets and graphs to describe patterns of change and solve problems.	Mission 6, Topic B Mission 6, Topic D
5.2.1.2	Use a rule or table to represent ordered pairs of positive integers and graph these ordered pairs on a coordinate system.	Mission 6, Topic B Mission 6, Topic D
5.2.2.1	Apply the commutative, associative and distributive properties and order of operations to generate equivalent numerical expressions and to solve problems involving whole numbers.	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 4, Topic D Mission 4, Topic G Mission 4, Topic H
5.2.3.1	Determine whether an equation or inequality involving a variable is true or false for a given value of the variable.	Grade 6, Mission 6, Topic A Grade 6, Mission 6, Topic B Grade 6, Mission 6, Topic C
5.2.3.2	Represent real-world situations using equations and inequalities involving variables. Create real-world situations corresponding to equations and inequalities.	Grade 6, Mission 6, Topic A Grade 6, Mission 6, Topic B
5.2.3.3	Evaluate expressions and solve equations involving variables when values for the variables are given.	Mission 5, Topic B Grade 6, Mission 6, Topic A Grade 6, Mission 6, Topic B Grade 6, Mission 6, Topic C
Geometry & Measurement		
5.3.1.1	Describe and classify three-dimensional figures including cubes, prisms and pyramids by the number of edges, faces or vertices as well as the types of faces.	Grade 6, Mission 1, Topic E *This topic focuses on describing and classifying prisms and pyramids by the number and type of faces. It is recommended that teachers use supplemental materials to address the depth of this standard.*

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
5th Grade Standards		**Lessons**
5.3.1.2	Recognize and draw a net for a three-dimensional figure.	Grade 6, Mission 1, Topic E Grade 6, Mission 1, Topic F Grade 6, Mission 1, Topic G
5.3.2.1	Develop and use formulas to determine the area of triangles, parallelograms and figures that can be decomposed into triangles.	Grade 6, Mission 1, Topic A Grade 6, Mission 1, Topic B Grade 6, Mission 1, Topic C Grade 6, Mission 1, Topic D Grade 6, Mission 1, Topic G
5.3.2.2	Use various tools and strategies to measure the volume and surface area of objects that are shaped like rectangular prisms.	Mission 5, Topic B Grade 6, Mission 1, Topic E
5.3.2.3	Understand that the volume of a three-dimensional figure can be found by counting the total number of same-sized cubic units that fill a shape without gaps or overlaps. Use cubic units to label volume measurements.	Mission 5, Topic A Mission 5, Topic B
5.3.2.4	Develop and use the formulas $V = \ell wh$ and $V = Bh$ to determine the volume of rectangular prisms. Justify why base area B and height h are multiplied to find the volume of a rectangular prism by breaking the prism into layers of unit cubes.	Mission 5, Topic B
Data Analysis		
5.4.1.1	Know and use the definitions of the mean, median and range of a set of data. Know how to use a spreadsheet to find the mean, median and range of a data set. Understand that the mean is a "leveling out" of data.	Grade 6, Mission 8, Topic C Grade 6, Mission 8, Topic D
5.4.1.1	Create and analyze double-bar graphs and line graphs by applying understanding of whole numbers, fractions and decimals. Know how to create spreadsheet tables and graphs to display data.	Mission 4, Topic A

6th Grade

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
Number & Operation		
6.1.1.1	Locate positive rational numbers on a number line and plot pairs of positive rational numbers on a coordinate grid.	Grade 5, Mission 6, Topic A Grade 5, Mission 6, Topic B Grade 5, Mission 6, Topic C Grade 5, Mission 6, Topic D
6.1.1.2	Compare positive rational numbers represented in various forms. Use the symbols <, = and >.	Grade 5, Mission 1, Topic B Grade 5, Mission 1, Topic D Grade 5, Mission 1, Topic E Grade 5, Mission 1, Topic F
6.1.1.3	Understand that percent represents parts out of 100 and ratios to 100.	Mission 3, Topic D
6.1.1.4	Determine equivalences among fractions, decimals and percents; select among these representations to solve problems.	Mission 3, Topic D
6.1.1.5	Factor whole numbers; express a whole number as a product of prime factors with exponents.	Mission 7, Topic D
6.1.1.6	Determine greatest common factors and least common multiples. Use common factors and common multiples to calculate with fractions and find equivalent fractions.	Mission 7, Topic D
6.1.1.7	Convert between equivalent representations of positive rational numbers.	Mission 7, Topic A
6.1.2.1	Identify and use ratios to compare quantities; understand that comparing quantities using ratios is not the same as comparing quantities using subtraction.	Mission 2, Topic A Mission 2, Topic B
6.1.2.2	Apply the relationship between ratios, equivalent fractions and percents to solve problems in various	Mission 3, Topic D Mission 3, Topic E

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
6th Grade Standards		**Lessons**
	contexts, including those involving mixtures and concentrations.	
6.1.2.3	Determine the rate for ratios of quantities with different units.	Mission 2, Topic C Mission 3, Topic A Mission 3, Topic C
6.1.2.4	Use reasoning about multiplication and division to solve ratio and rate problems.	Mission 2, Topic D Mission 2, Topic E Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C
6.1.3.1	Multiply and divide decimals and fractions, using efficient and generalizable procedures, including standard algorithms.	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D Mission 4, Topic E Mission 5, Topic C Mission 5, Topic D
6.1.3.2	Use the meanings of fractions, multiplication, division and the inverse relationship between multiplication and division to make sense of procedures for multiplying and dividing fractions.	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D Mission 4, Topic E
6.1.3.3	Calculate the percent of a number and determine what percent one number is of another number to solve problems in various contexts.	Mission 3, Topic D
6.1.3.4	Solve real-world and mathematical problems requiring arithmetic with decimals, fractions and mixed numbers.	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D Mission 4, Topic E Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E Mission 6, Topic A Mission 8, Topic C

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
6.1.3.5	Estimate solutions to problems with whole numbers, fractions and decimals and use the estimates to assess the reasonableness of results in the context of the problem.	Grade 7, Mission 6, Topic A Grade 7, Mission 6, Topic B
Algebra		
6.2.1.1	Understand that a variable can be used to represent a quantity that can change, often in relationship to another changing quantity. Use variables in various contexts.	Mission 6, Topic A Mission 6, Topic B Mission 7, Topic B
6.2.1.2	Represent the relationship between two varying quantities with function rules, graphs and tables; translate between any two of these representations.	Mission 6, Topic D
6.2.2.1	Apply the associative, commutative and distributive properties and order of operations to generate equivalent expressions and to solve problems involving positive rational numbers.	Mission 1, Topic F Mission 6, Topic C
6.2.3.1	Represent real-world or mathematical situations using equations and inequalities involving variables and positive rational numbers.	Mission 6, Topic A Mission 6, Topic B Mission 7, Topic B
6.2.3.2	Solve equations involving positive rational numbers using number sense, properties of arithmetic and the idea of maintaining equality on both sides of the equation. Interpret a solution in the original context and assess the reasonableness of results.	Mission 6, Topic A Mission 6, Topic B
Geometry & Measurement		
6.3.1.1	Calculate the surface area and volume of prisms and use appropriate units, such as cm2 and cm3. Justify the formulas used. Justification may involve decomposition, nets or other models.	Mission 1, Topic E Mission 1, Topic F Mission 1, Topic G Mission 4, Topic D Mission 4, Topic E
6.3.1.2	Calculate the area of quadrilaterals. Quadrilaterals include squares, rectangles, rhombuses, parallelograms, trapezoids and kites. When formulas are used, be able to explain why they are valid.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
6th Grade Standards		**Lessons**
		Mission 1, Topic G Mission 4, Topic D
6.3.1.3	Estimate the perimeter and area of irregular figures on a grid when they cannot be decomposed into common figures and use correct units, such as *cm* and cm^2.	Mission 1, Topic A
6.3.2.1	Solve problems using the relationships between the angles formed by intersecting lines.	Grade 7, Mission 7, Topic A
6.3.2.2	Determine missing angle measures in a triangle using the fact that the sum of the interior angles of a triangle is 180˚. Use models of triangles to illustrate this fact.	Grade 8, Mission 1, Topic D
6.3.2.3	Develop and use formulas for the sums of the interior angles of polygons by decomposing them into triangles.	*It is recommended that teachers use supplemental materials to fully address this standard.*
6.3.3.1	Solve problems in various contexts involving conversion of weights, capacities, geometric measurements and times within measurement systems using appropriate units.	Grade 5, Mission 1, Topic A Grade 5, Mission 2, Topic D Grade 5, Mission 4, Topic C Grade 5, Mission 4, Topic E
6.3.3.2	Estimate weights, capacities and geometric measurements using benchmarks in measurement systems with appropriate units.	Grade 5, Mission 1, Topic A Grade 5, Mission 2, Topic D Grade 5, Mission 4, Topic C Grade 5, Mission 4, Topic E
Data Analysis & Probability		
6.4.1.1	Determine the sample space (set of possible outcomes) for a given experiment and determine which members of the sample space are related to certain events. Sample space may be determined by the use of tree diagrams, tables or pictorial representations.	Grade 7, Mission 8, Topic A Grade 7, Mission 8, Topic B
6.4.1.2	Determine the probability of an event using the ratio between the size of the event and the size of the sample space; represent probabilities as percents, fractions and decimals between 0 and 1 inclusive. Understand that probabilities measure likelihood.	Grade 7, Mission 8, Topic A Grade 7, Mission 8, Topic B

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
6th Grade Standards		**Lessons**
6.4.1.3	Perform experiments for situations in which the probabilities are known, compare the resulting relative frequencies with the known probabilities; know that there may be differences.	Grade 7, Mission 8, Topic A Grade 7, Mission 8, Topic B
6.4.1.4	Calculate experimental probabilities from experiments; represent them as percents, fractions and decimals between 0 and 1 inclusive. Use experimental probabilities to make predictions when actual probabilities are unknown.	Grade 7, Mission 8, Topic A Grade 7, Mission 8, Topic B

7th Grade

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
Number & Operation		
7.1.1.1	Know that every rational number can be written as the ratio of two integers or as a terminating or repeating decimal. Recognize that π is not rational, but that it can be approximated by rational numbers such as 22/7 and 3.14.	Grade 8, Mission 8, Topic E
7.1.1.2	Understand that division of two integers will always result in a rational number. Use this information to interpret the decimal result of a division problem when using a calculator.	Grade 8, Mission 8, Topic E
7.1.1.3	Locate positive and negative rational numbers on a number line, understand the concept of opposites, and plot pairs of positive and negative rational numbers on a coordinate grid.	Grade 6, Mission 7, Topic A Grade 6, Mission 7, Topic C
7.1.1.4	Compare positive and negative rational numbers expressed in various forms using the symbols < , > , = , ≤ , ≥ .	Grade 6, Mission 7, Topic A
7.1.1.5	Recognize and generate equivalent representations of positive and negative rational numbers, including equivalent fractions.	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E Mission 5, Topic F
7.1.2.1	Add, subtract, multiply and divide positive and negative rational numbers that are integers, fractions and terminating decimals; use efficient and generalizable procedures, including standard algorithms; raise positive rational numbers to whole-number exponents.	Mission 4, Topic A Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E Mission 5, Topic F Mission 6, Topic D Mission 7, Topic B Mission 8, Topic D

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
7.1.2.2	Use real-world contexts and the inverse relationship between addition and subtraction to explain why the procedures of arithmetic with negative rational numbers make sense.	Mission 5, Topic A Mission 5, Topic B Mission 6, Topic D Mission 7, Topic B
7.1.2.3	Understand that calculators and other computing technologies often truncate or round numbers.	*It is recommended that teachers use supplemental materials to fully address this standard.*
7.1.2.4	Solve problems in various contexts involving calculations with positive and negative rational numbers and positive integer exponents, including computing simple and compound interest.	Mission 3, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E Mission 5, Topic F Grade 8, Mission 7, Topic B Grade 8, Mission 7, Topic C
7.1.2.5	Use proportional reasoning to solve problems involving ratios in various contexts.	Mission 3, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D
7.1.2.6	Demonstrate an understanding of the relationship between the absolute value of a rational number and distance on a number line. Use the symbol for absolute value.	Grade 6, Mission 7, Topic A
Algebra		
7.2.1.1	Understand that a relationship between two variables, x and y, is proportional if it can be expressed in the form $y/x = k$ or $y=kx$. Distinguish proportional relationships from other relationships, including inversely proportional relationships ($xy=k$ or $y=k/x$).	Mission 2, Topic B Mission 2, Topic C
7.2.1.2	Understand that the graph of a proportional relationship is a line through the origin whose slope is the unit rate (constant of proportionality). Know how	Mission 2, Topic D

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
7th Grade Standards		**Lessons**
	to use graphing technology to examine what happens to a line when the unit rate is changed.	
7.2.2.1	Represent proportional relationships with tables, verbal descriptions, symbols, equations and graphs; translate from one representation to another. Determine the unit rate (constant of proportionality or slope) given any of these representations.	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E
7.2.2.2	Solve multi-step problems involving proportional relationships in numerous contexts.	Mission 3, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D
7.2.2.3	Use knowledge of proportions to assess the reasonableness of solutions.	Mission 6, Topic B
7.2.2.4	Represent real-world or mathematical situations using equations and inequalities involving variables and positive and negative rational numbers	Mission 5, Topic E Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C Mission 7, Topic A
7.2.3.1	Use properties of algebra to generate equivalent numerical and algebraic expressions containing rational numbers, grouping symbols and whole number exponents. Properties of algebra include associative, commutative and distributive laws.	Mission 6, Topic D
7.2.3.2	Evaluate algebraic expressions containing rational numbers and whole number exponents at specified values of their variables.	Grade 6, Mission 1, Topic B Grade 6, Mission 1, Topic C Grade 6, Mission 6, Topic B Grade 6, Mission 6, Topic C
7.2.3.3	Apply understanding of order of operations and grouping symbols when using calculators and other technologies.	Grade 6, Mission 1, Topic B Grade 6, Mission 1, Topic C Grade 6, Mission 6, Topic B Grade 6, Mission 6, Topic C
7.2.4.1	Represent relationships in various contexts with equations involving variables and positive and negative rational numbers. Use the properties of	Mission 6, Topic A Mission 6, Topic B

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
7th Grade Standards		**Lessons**
	equality to solve for the value of a variable. Interpret the solution in the original context.	
7.2.4.2	Solve equations resulting from proportional relationships in various contexts.	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D
Geometry & Measurement		
7.3.1.1	Demonstrate an understanding of the proportional relationship between the diameter and circumference of a circle and that the unit rate (constant of proportionality) is π. Calculate the circumference and area of circles and sectors of circles to solve problems in various contexts.	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C
7.3.1.2	Calculate the volume and surface area of cylinders and justify the formulas used.	Grade 8, Mission 5, Topic D
7.3.2.1	Describe the properties of similarity, compare geometric figures for similarity, and determine scale factors.	Grade 8, Mission 2, Topic B
7.3.2.2	Apply scale factors, length ratios and area ratios to determine side lengths and areas of similar geometric figures.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C
7.3.2.3	Use proportions and ratios to solve problems involving scale drawings and conversions of measurement units.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 2, Topic A Mission 3, Topic B Mission 3, Topic C
7.3.2.4	Graph and describe translations and reflections of figures on a coordinate grid and determine the coordinates of the vertices of the figure after the transformation.	Grade 8, Mission 1, Topic A
Data Analysis & Probability		

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
7.4.1.1	Design simple experiments and collect data. Determine mean, median and range for quantitative data and from data represented in a display. Use these quantities to draw conclusions about the data, compare different data sets, and make predictions.	Mission 8, Topic A Mission 8, Topic B Mission 8, Topic C Mission 8, Topic D Mission 8, Topic E
7.4.1.2	Describe the impact that inserting or deleting a data point has on the mean and the median of a data set. Know how to create data displays using a spreadsheet to examine this impact.	Mission 8, Topic C Mission 8, Topic D
7.4.2.1	Use reasoning with proportions to display and interpret data in circle graphs (pie charts) and histograms. Choose the appropriate data display and know how to create the display using a spreadsheet or other graphing technology.	Grade 6, Mission 8, Topic B
7.4.3.1	Use random numbers generated by a calculator or a spreadsheet or taken from a table to simulate situations involving randomness, make a histogram to display the results, and compare the results to known probabilities.	Mission 8, Topic A
7.4.3.2	Calculate probability as a fraction of sample space or as a fraction of area. Express probabilities as percents, decimals and fractions.	Mission 8, Topic A
7.4.3.3	Use proportional reasoning to draw conclusions about and predict relative frequencies of outcomes based on probabilities.	Mission 8, Topic A Mission 8, Topic C Mission 8, Topic E

8th Grade

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
8th Grade Standards		Lessons
Number & Operation		
8.1.1.1	Classify real numbers as rational or irrational. Know that when a square root of a positive integer is not an integer, then it is irrational. Know that the sum of a rational number and an irrational number is irrational, and the product of a non-zero rational number and an irrational number is irrational.	Mission 8, Topic E
8.1.1.2	Compare real numbers; locate real numbers on a number line. Identify the square root of a positive integer as an integer, or if it is not an integer, locate it as a real number between two consecutive positive integers.	Mission 8, Topic A Mission 8, Topic B Mission 8, Topic C Mission 8, Topic D Mission 8, Topic E
8.1.1.3	Determine rational approximations for solutions to problems involving real numbers.	Mission 8, Topic A Mission 8, Topic B Mission 8, Topic C Mission 8, Topic D Mission 8, Topic E
8.1.1.4	Know and apply the properties of positive and negative integer exponents to generate equivalent numerical expressions.	Mission 7, Topic B Mission 7, Topic C
8.1.1.5	Express approximations of very large and very small numbers using scientific notation; understand how calculators display numbers in scientific notation. Multiply and divide numbers expressed in scientific notation, express the answer in scientific notation, using the correct number of significant digits when physical measurements are involved.	Mission 7, Topic C Mission 7, Topic D
Algebra		
8.2.1.1	Understand that a function is a relationship between an independent variable and a dependent variable in which the value of the independent variable determines the value of the dependent variable. Use	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic E

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
8th Grade Standards		**Lessons**
	functional notation, such as f(x), to represent such relationships.	
8.2.1.2	Use linear functions to represent relationships in which changing the input variable by some amount leads to a change in the output variable that is a constant times that amount.	Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E
8.2.1.3	Understand that a function is linear if it can be expressed in the form or if its graph is a straight line.	Mission 5, Topic B Mission 5, Topic C Mission 5, Topic E
8.2.1.4	Understand that an arithmetic sequence is a linear function that can be expressed in the form , where $f(x) = m + b$, where $x = 0, 1, 2, 3, \ldots$	*It is recommended that teachers use supplemental materials to fully address this standard.*
8.2.1.5	Understand that a geometric sequence is a non-linear function that can be expressed in the form $f(x) = ab^x$, where $x = 0, 1, 2, 3, \ldots.$	*It is recommended that teachers use supplemental materials to fully address this standard.*
8.2.2.1	Represent linear functions with tables, verbal descriptions, symbols, equations and graphs; translate from one representation to another.	Mission 5, Topic B Mission 5, Topic C
8.2.2.2	Identify graphical properties of linear functions including slopes and intercepts. Know that the slope equals the rate of change, and that the yintercept is zero when the function represents a proportional relationship.	Mission 2, Topic C Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic E
8.2.2.3	Identify how coefficient changes in the equation $f(x) = m + b$ affect the graphs of linear functions. Know how to use graphing technology to examine these effects.	Mission 5, Topic C
8.2.2.4	Represent arithmetic sequences using equations, tables, graphs and verbal descriptions, and use them to solve problems.	*It is recommended that teachers use supplemental materials to fully address this standard.*
8.2.2.5	Represent geometric sequences using equations, tables, graphs and verbal descriptions, and use them to solve problems.	*It is recommended that teachers use supplemental materials to fully address this standard.*

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
8th Grade Standards		**Lessons**
8.2.3.1	Evaluate algebraic expressions, including expressions containing radicals and absolute values, at specified values of their variables.	*It is recommended that teachers use supplemental materials to fully address this standard.*
8.2.3.2	Justify steps in generating equivalent expressions by identifying the properties used, including the properties of algebra. Properties include the associative, commutative and distributive laws, and the order of operations, including grouping symbols.	*It is recommended that teachers use supplemental materials to fully address this standard.*
8.2.4.1	Use linear equations to represent situations involving a constant rate of change, including proportional and non-proportional relationships.	Mission 3, Topic A Mission 3, Topic B
8.2.4.2	Solve multi-step equations in one variable. Solve for one variable in a multi-variable equation in terms of the other variables. Justify the steps by identifying the properties of equalities used.	Mission 4, Topic B
8.2.4.3	Express linear equations in slope-intercept, point-slope and standard forms, and convert between these forms. Given sufficient information, find an equation of a line.	Mission 3, Topic C Mission 3, Topic D Mission 5, Topic A
8.2.4.4	Use linear inequalities to represent relationships in various contexts.	Grade 7, Mission 6, Topic C
8.2.4.5	Solve linear inequalities using properties of inequalities. Graph the solutions on a number line.	Grade 7, Mission 6, Topic C
8.2.4.6	Represent relationships in various contexts with equations and inequalities involving the absolute value of a linear expression. Solve such equations and inequalities and graph the solutions on a number line.	*It is recommended that teachers use supplemental materials to fully address this standard.*
8.2.4.7	Represent relationships in various contexts using systems of linear equations. Solve systems of linear equations in two variables symbolically, graphically and numerically.	Mission 3, Topic D Mission 3, Topic E Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D
8.2.4.8	Understand that a system of linear equations may have no solution, one solution, or an infinite number	Mission 3, Topic D Mission 3, Topic E

MINNESOTA'S ACADEMIC STANDARDS IN MATHEMATICS		ZEARN MATH
8th Grade Standards		**Lessons**
	of solutions. Relate the number of solutions to pairs of lines that are intersecting, parallel or identical. Check whether a pair of numbers satisfies a system of two linear equations in two unknowns by substituting the numbers into both equations.	Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D
8.2.4.9	Use the relationship between square roots and squares of a number to solve problems.	Mission 8, Topic B Mission 8, Topic C Mission 8, Topic D
Geometry & Measurement		
8.3.1.1	Use the Pythagorean Theorem to solve problems involving right triangles.	Mission 8, Topic C
8.3.1.2	Determine the distance between two points on a horizontal or vertical line in a coordinate system. Use the Pythagorean Theorem to find the distance between any two points in a coordinate system.	Mission 8, Topic C
8.3.1.3	Informally justify the Pythagorean Theorem by using measurements, diagrams and computer software.	Mission 8, Topic C
8.3.2.1	Understand and apply the relationships between the slopes of parallel lines and between the slopes of perpendicular lines. Dynamic graphing software may be used to examine these relationships.	*It is recommended that teachers use supplemental materials to fully address this standard.*
8.3.2.2	Analyze polygons on a coordinate system by determining the slopes of their sides.	*It is recommended that teachers use supplemental materials to fully address this standard.*
8.3.2.3	Given a line on a coordinate system and the coordinates of a point not on the line, find lines through that point that are parallel and perpendicular to the given line, symbolically and graphically.	*It is recommended that teachers use supplemental materials to fully address this standard.*
Data Analysis & Probability		
8.4.1.1	Collect, display and interpret data using scatterplots. Use the shape of the scatterplot to informally estimate a line of best fit and determine an equation for the	Mission 6, Topic A Mission 6, Topic B

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	line. Use appropriate titles, labels and units. Know how to use graphing technology to display scatterplots and corresponding lines of best fit.	
8.4.1.2	Use a line of best fit to make statements about approximate rate of change and to make predictions about values not in the original data set.	Mission 6, Topic A Mission 6, Topic B
8.4.1.3	Assess the reasonableness of predictions using scatterplots by interpreting them in the original context.	Mission 6, Topic A Mission 6, Topic B